

**Acceptable Means of Compliance and Guidance Material to Annex I (Part-FCL) to
Regulation (EU) No 1178/2011
Issue 1, Amendment 15**

Annex II to ED Decision 2026/006/R

‘AMC and GM to Part-FCL — Issue 1, Amendment 15’

The amendments are presented as follows:

- deleted text is ~~struck through~~;
- new text is highlighted;
- an ellipsis ‘[...]’ indicates that the rest of the text is unchanged.

Note to readers

‘Agency’ and ‘EASA’ are used interchangeably, and this is most visible in consolidated texts. In all cases, both terms refer to the European Union Aviation Safety Agency.

The Annex to Decision 2011/016/R of the Executive Director of the Agency of 15 December 2011 (AMC & GM to Part-FCL — Initial issue) is amended as follows:

GM1 FCL.036(b)(1) Use of FSTDs for training, testing and checking

TACTILE HARDWARE AND TOUCHSCREEN REPRESENTATION

In CS-FSTD Issue 1, the term ‘tactile hardware’ is used to refer to the flight deck layout and structure feature of an FSTD in order to set out the flight deck haptic requirements of FSTDs whose flight deck instrument, panels and switches are made up of physical hardware components.

CS-FSTD Issue 1 also allows the possibility to qualify FSTDs whose flight deck layout and structure feature, for fidelity levels R and G, may represent flight deck instrument, panels and switches in the form of touchscreens. Tactile hardware and touchscreen representation terminologies are used in opposition.

In the context of the use of FSTDs for training, testing and checking, in general only FSTDs with tactile hardware should be used. FSTDs with touchscreen representations may be used in type rating training course for the conduct of procedure training on normal, abnormal and emergency systems in accordance with the arrangements contained in AMC1 to Appendix 9 Section A point 1f.

AMC2 FCL.110.H LAPL(H) Experience requirements and crediting

USE OF FSTDs IN LAPL(H) TRAINING

(a) If an FSTD is used in accordance with the provisions of point FCL.110.H(aa), it should:

- (1) be qualified with an FCS or an assigned FCS in its FSTD qualification certificate;
- (2) have all the following features at a fidelity level that is equal to, or higher than, R (representative):
 - (i) Flight deck layout and structure;
 - (ii) Flight controls forces and hardware;
 - (iii) Flight controls systems operation;
 - (iv) Aircraft systems;
 - (v) Performance and handling on ground;
 - (vi) Performance and handling in ground effect;
 - (vii) Performance and handling out of ground effect; and
- (3) additionally, for those training tasks requiring cueing or environment simulation features, have an adequate fidelity level for the relevant features to complete such training tasks. The

determination of the adequate fidelity levels should be based on an analysis of the training needs, taking into consideration the general requirements of CS-FSTD.

- (b) Legacy FSTDs may be used for training in accordance with the provisions of point FCL.110.H(aa), provided that the training organisation demonstrates that the FSTD represents the type of helicopter used for the skill test, taking into consideration the technical specifications of the training device included in the qualification certificate and in the equipment specification list (ESL).

AMC1 FCL.210.H PPL(H) Experience requirements and crediting

USE OF FSTDs IN PPL(H) TRAINING

The use of FSTDs in accordance with point FCL.210.H(aa)(2) should follow the content of AMC2 FCL.110.H.

AMC1 FCL.930.TRI TRI — Training course

TRI TRAINING COURSE — AEROPLANES

[...]

- (b) Content

[...]

- (3) Part 3 — Flight instruction

[...]

- (ii) Use of FSTDs

[...]

~~(D) — Where no FSTD exists for the type of aeroplane for which the certificate is sought, or if the FSTD is not suitable to complete all the elements of the training programme for the TRI certificate, the entire course or a part of it should be conducted in the applicable aeroplane type, and the synthetic device elements should be replaced with appropriate exercises in the aeroplane.~~

~~The assessment of competence should be performed:~~

~~— when no FSTD exists, in the aeroplane; and~~

~~— when not all elements of the training are completed in the FSTD, in both the aeroplane and the FSTD; this combined use of aeroplane and FSTD in the assessment of competence should reflect and be similar to the combined use of aeroplane and FSTD during the training course.~~

- (F) (D) In general, TRI training is designed to develop the competencies of a pilot to become an instructor. From this perspective, the training may be provided in several arrangements:

[...]

AMC2 FCL.930.TRI TRI — Training course

HELICOPTERS

[...]

Part 3

FLIGHT INSTRUCTION SYLLABUS

[...]

~~TRAINING WHERE NO FSTD EXISTS~~

~~(m) Where no FSTD exists for the type for which the TRI(H) certificate is sought, a similar course of training should be conducted in the applicable helicopter type. This includes all elements listed under sub paragraphs (k)(1) and (2) of this AMC, the FSTD elements being replaced with appropriate exercises in a helicopter of the applicable type, subject to any restrictions placed on the conduct of critical exercises associated with helicopter flight manual limitations and safety considerations.~~

SPECIFIC TRAINING: LANDING TRAINING

(n) (m) [...]

AMC2 FCL.1015 Examiner standardisation

[...]

- (p) Before undertaking a test or check, an examiner ~~will~~ **should** verify that the aircraft or FSTD intended to be used is suitable and appropriately equipped for the test or check. **In the case of FSTDs, such verification should be based on the technical specifications of the training device included in the FSTD qualification certificate and in the ESL.** Aircraft that fall under points (a), (b), (c), or (d) of Annex I to the Basic Regulation can be used provided that they are subject to an authorisation as per point ORA.ATO.135 or point DTO.GEN.240.

[...]

AMC1 to Appendix 9 Section A point 1f Training, skill test and proficiency check for the MPL, and the ATPL, and for type and class ratings, and proficiency checks for the BIR and the IR

TRAINING MATRICES FOR TYPE RATING TRAINING

(a) General

- (1) The training matrices should be used to support the design of a type rating training course in accordance with the task-to-tool methodology described in AMC5 ORA.ATO.125.
- (2) This AMC illustrates, for each training task, the minimum fidelity levels that, for each of its features, an FSTD with an FCS or assigned FCS should have.

For each training task, the following is given:
 - (i) T (Training): The minimum FCS or assigned FCS that an FSTD should have for the introduction of a training task. In such an FSTD, the relevant training task may be commenced and practised during a type rating course in accordance with Part-FCL, but it should not be completed to proficiency for the purpose of issuing or renewing a type rating.
 - (ii) TP (Training to Proficiency): The minimum FCS or assigned FCS that an FSTD should have for the completion of a training task. To complete the relevant training task to proficiency, as required by Appendix 9 to Part-FCL for the purpose of issuing or renewing a type rating, such an FSTD should be used.
- (3) When, in accordance with AMC5 ORA.ATO.125, determining the suitability of an FSTD with an FCS or assigned FCS for the introduction, or completion to proficiency, of training tasks, the following training matrices should be used:

(i) for multi-pilot aeroplanes and for single-pilot high-performance complex aeroplanes:

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight control forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
Section 1 — Flight preparation																
1.1	Performance calculation	TP	I	I	I	I	I	I	I	I	I	I	I	I	I	I
		T	I	I	I	I	I	I	I	I	I	I	I	I	I	I
1.2	Aeroplane external visual inspection; location of each item and purpose of inspection	TP	I	I	I	I	I	I	I	I	I	I	I	I	I	I
		T	I	I	I	I	I	I	I	I	I	I	I	I	I	I
1.3	Cockpit inspection	TP	S	R	G	S	G	N	N	N	N	N	N	N	G	N
		T	R	R	G	R	G	N	N	N	N	N	N	N	G	N
1.4	Use of checklist prior to starting engines, starting procedures, radio and navigation equipment check, selection and setting of navigation and communication frequencies	TP	S	R	G	S	G	N	N	R	R	G	G	S	R	G
		T	G	N	G	R	G	N	N	G	N	N	N	R	G	N
1.5	Taxiing in compliance with ATC instructions or instructions of instructor	TP	S	S	S	S	S	N	N	R	R	S	R	S	R	G
		T	G	R	R	R	R	N	N	G	N	N	R	R	G	G
1.6	Before take-off checks	TP	S	S	S	S	R	N	N	R	R	R	R	S	R	G
		T	G	G	G	R	G	N	N	G	N	N	N	R	G	N
Section 2 — Take-offs																
2.1	Normal take-offs with different flap settings, including expedited take-off	TP	S	S	S	S	S	S	S	R	R	S	S	N	R	R
		T	R	R	R	R	R	R	R	G	N	N	R	N	G	G
2.2		TP	S	S	S	S	S	S	S	R	R	S	S	S	R	R

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight control forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
	Instrument take-off; transition to instrument flight is required during rotation or immediately after becoming airborne	T	R	R	R	R	R	R	R	G	N	N	R	R	G	G
2.3	Crosswind take-off	TP	S	S	S	S	S	S	S	R	R	S	S	N	R	R
		T	R	R	R	R	R	R	R	G	N	R	R	N	G	G
2.4	Take-off at maximum take-off mass (actual or simulated maximum take-off mass)	TP	S	S	S	S	S	S	S	R	R	S	S	N	R	R
		T	R	R	R	R	R	R	R	G	N	N	R	N	G	G
2.5.1	Take-offs with simulated engine failure shortly after reaching V2	TP	S	S	S	S	S	S	S	R	R	S	S	N	R	R
		T	R	R	R	R	R	R	R	G	G	R	R	N	G	G
2.5.2	Take-offs with simulated engine failure between V1 and V2	TP	S	S	S	S	S	S	S	R	R	S	S	S	S	S
		T	R	R	R	R	R	R	R	G	G	R	R	N	G	G
2.6	Rejected take-off at a reasonable speed before reaching V1	TP	S	S	S	S	S	N	N	R	R	S	S	N	R	R
		T	R	R	R	R	R	N	N	G	G	R	R	N	G	G
Section 3 – Flight manoeuvres and procedures																
3.1	Manual flight with and without flight directors (no autopilot, no autothrust/autothrottle, and at different control laws, where applicable)	TP	R	S	S	S	N	N	S	G	R	R	R	N	G	N
		T	G	R	R	R	N	N	R	N	N	N	G	N	G	N
3.1.1	At different speeds (including slow flight) and altitudes within the FSTD training envelope	TP	R	S	S	S	N	N	S	G	R	R	R	N	G	N
		T	G	R	R	R	N	N	R	N	N	N	G	N	G	N
3.1.2	Steep turns using 45° bank, 180° to 360° left and right	TP	R	S	S	S	N	N	S	G	R	R	R	N	G	N
		T	G	R	R	R	N	N	R	N	N	N	G	N	G	N
3.1.3	Turns with and without spoilers	TP	R	S	S	S	N	N	S	G	R	R	R	N	G	N
		T	G	R	R	R	N	N	R	N	N	N	G	N	G	N

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight control forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
3.1.4	Procedural instrument flying and manoeuvring including instrument departure and arrival, and visual approach	TP	S	S	S	S	N	N	S	G	R	R	R	S	G	G
		T	G	R	R	R	N	N	R	N	N	N	G	R	G	G
3.2	Tuck under and Mach buffets (if applicable), and other specific flight characteristics of the aeroplane (e.g. Dutch Roll)	TP	S	S	S	S	S	S	S	G	S	S	S	S	S	S
		T	G	R	R	R	N	N	R	N	G	G	G	N	G	N
3.3	Normal operation of systems and controls engineer's panel (if applicable)	TP	R	N	N	S	N	N	N	N	N	N	N	N	G	N
		T	G	N	N	R	N	N	N	N	N	N	N	N	G	N
Section 3.4 – Normal and abnormal operations of following systems																
3.4.0	Engine (if necessary, propeller)	TP	S	S	S	S	N	N	S	R	R	R	N	N	G	N
		T	G	G	G	R	N	N	G	G	N	N	N	N	G	N
3.4.1	Pressurisation and air conditioning	TP	R	N	G	S	N	N	G	N	N	N	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
3.4.2	Pitot/static system	TP	R	N	G	S	N	N	G	N	N	N	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
3.4.3	Fuel system	TP	R	N	G	S	N	N	G	N	N	N	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
3.4.4	Electrical system	TP	R	N	G	S	N	N	G	N	N	N	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
3.4.5	Hydraulic system	TP	R	N	G	S	N	N	G	N	N	N	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
3.4.6	Flight control and trim system	TP	R	S	S	S	N	N	S	N	R	R	N	N	G	N
		T	G	R	R	R	N	N	G	N	N	N	N	N	G	N
3.4.7		TP	R	N	G	S	N	N	S	N	R	R	R	N	R	N

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight control forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
	Anti-icing/de-icing system, glare shield heating	T	G	N	G	R	N	N	G	N	N	N	N	N	R	N
3.4.8	Autopilot/flight director	TP	S	R	R	S	N	N	S	N	N	N	N	R	G	N
		T	G	G	G	R	N	N	G	N	N	N	N	R	G	N
3.4.9	Stall warning devices or stall avoidance devices, and stability augmentation devices	TP	R	R	R	S	N	N	S	N	N	N	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
3.4.10	Ground proximity warning system, weather radar, radio altimeter, transponder	TP	R	R	R	S	N	N	S	N	N	N	N	N	R	R
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	R
3.4.11	Radios, navigation equipment, instruments, FMS	TP	R	N	G	S	N	N	G	N	N	N	N	S	R	G
		T	G	N	G	R	N	N	G	N	N	N	N	S	G	G
3.4.12	Landing gear and brake	TP	R	S	S	S	S	S	S	R	R	R	G	N	G	G
		T	G	R	R	R	R	R	R	G	N	N	N	N	G	N
3.4.13	Slat and flap system	TP	R	S	S	S	S	S	S	N	N	R	N	N	G	N
		T	G	R	R	R	R	R	R	N	N	N	N	N	G	N
3.4.14	Auxiliary power unit (APU)	TP	R	N	G	S	N	N	G	N	N	N	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
Section 3.6 — Abnormal and emergency procedures																
3.6.1	Fire drills e.g. engine, APU, cabin, cargo compartment, flight deck, wing and electrical fires including evacuation	TP	R	N	G	S	N	N	G	N	R	R	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
3.6.2	Smoke control and removal	TP	R	N	G	S	N	N	G	N	N	N	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
3.6.3	Engine failures, shutdown and restart at a safe height	TP	S	S	S	S	N	N	S	R	R	R	N	N	G	N
		T	G	G	G	R	N	N	G	G	N	N	N	N	G	N

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight control forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
3.6.4	Fuel dumping (simulated)	TP	R	N	G	S	N	N	G	N	N	N	N	N	G	N
		T	G	N	G	R	N	N	G	N	N	N	N	N	G	N
3.6.5	Wind shear at take-off/landing	TP	S	S	S	S	S	S	S	R	R	S	S	S	S	S
		T	R	R	R	R	R	R	R	G	N	N	R	N	R	G
3.6.6	Simulated cabin pressure failure/emergency descent	TP	S	R	R	S	N	N	S	G	R	R	N	N	G	R
		T	G	G	G	R	N	N	R	N	N	N	N	N	G	N
3.6.7	Incapacitation of flight crew member	TP	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		T	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3.6.8	Other emergency procedures as outlined in the appropriate aeroplane flight manual (AFM)	TP	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		T	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3.6.9	TCAS event	TP	S	S	S	S	S	S	S	R	R	S	S	S	S	S
		T	G	R	R	R	N	N	G	N	N	N	N	N	G	N
Section 3.7 — Upset prevention and recovery																
3.7.1	Recovery from stall events in: — take-off configuration — clean configuration at low altitude — clean configuration near maximum operating altitude — landing configuration	TP	S	S	S	S	S	S	S	R	S	S	S	S	S	S
		T	S	S	S	S	S	S	S	R	S	S	S	S	S	S
3.7.2	The following upset exercises: — recovery from nose-high at various bank angles; and — recovery from nose-low at various bank angles	TP	S	S	S	S	S	S	S	R	S	S	S	S	S	S
		T	S	S	S	S	S	S	S	R	S	S	S	S	S	S

Manoeuvres/Procedures		Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight control forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
Section 3.8 — Instrument flight procedures																
3.8.1	Adherence to departure and arrival routes and ATC instructions	TP	R	R	R	S	N	N	R	G	N	N	N	S	R	N
		T	G	G	G	R	N	N	G	N	N	N	N	R	G	N
3.8.2	Holding procedures	TP	R	R	R	S	N	N	R	G	N	N	N	S	R	N
		T	G	G	G	R	N	N	G	N	N	N	N	R	G	N
3.8.3.1	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach procedure manually, without flight director	TP	S	S	S	S	N	N	S	R	N	R	R	S	R	G
		T	G	R	R	R	N	N	G	G	N	N	G	R	G	G
3.8.3.2	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach procedure manually, with flight director	TP	S	S	S	S	N	N	S	R	N	R	R	S	R	G
		T	G	R	R	R	N	N	G	G	N	N	G	R	G	G
3.8.3.3	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach procedure with autopilot	TP	S	S	S	S	N	N	S	R	N	N	R	S	R	G
		T	G	G	G	R	N	N	G	G	N	N	G	R	G	G
3.8.3.4	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach procedure manually, with one engine simulated inoperative during final approach, either until touchdown or through the complete missed approach procedure (as applicable), starting: (i) before passing 1 000 ft above aerodrome level; and (ii) after passing 1 000 ft above aerodrome level.	TP	S	S	S	S	S	S	S	R	R	R	S	S	R	R
		T	G	R	R	R	G	G	G	G	N	N	G	R	G	G
3.8.4	2D operations down to the MDH/A	TP	S	S	S	S	N	N	S	R	N	R	R	S	R	G
		T	G	G	G	R	N	N	G	G	N	N	G	R	G	G

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight control forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
3.8.5	Circling approach under the following conditions: (a)*approach to the authorised minimum circling approach altitude at the aerodrome in question in accordance with the local instrument approach facilities in simulated instrument flight conditions; followed by: (b) circling approach to another runway at least 90° off centreline from the final approach used in item (a), at the authorised minimum circling approach altitude.	TP	S	S	S	S	N	N	S	R	N	R	S	S	R	R
		T	R	R	R	R	N	N	G	G	N	N	R	R	G	R
3.8.6	Visual approaches	TP	S	S	S	S	N	N	S	R	N	R	S	N	R	R
		T	R	R	R	R	N	N	G	G	N	N	R	N	G	R
Section 4 — Missed approach procedures																
4.1	Go-around with all engines operating during a 3D operation on reaching DH	TP	S	S	S	S	N	N	S	R	N	S	N	S	R	G
		T	R	R	R	R	N	N	R	G	N	N	N	R	G	G
4.2	Go-around with all engines operating from various stages during an instrument approach	TP	S	S	S	S	N	N	S	R	N	S	N	S	R	G
		T	R	R	R	R	N	N	R	G	N	N	N	R	G	G
4.3	Other missed approach procedures	TP	S	S	S	S	N	N	S	R	N	S	N	S	R	G
		T	R	R	R	R	N	N	R	G	N	N	N	R	G	G
4.4	Manual go-around with the critical engine simulated inoperative after an instrument approach on reaching DH, MDH or MAPt	TP	S	S	S	S	N	N	S	R	R	S	N	S	R	G
		T	R	R	R	R	N	N	R	G	N	N	N	R	G	G
4.5	Rejected landing with all engines operating:	TP	S	S	S	S	S	S	S	R	R	S	S	S	R	R

Manoeuvres/Procedures		Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight control forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
	— from various heights below DH/MDH; — after touchdown (balked landing)	T	R	R	R	R	R	R	R	G	N	N	G	R	G	R
Section 5 – Landings																
5.1	Normal landings with visual reference established when reaching DA/H following an instrument approach operation	TP	S	S	S	S	S	S	S	R	R	S	S	S	R	R
		T	R	R	R	R	R	R	R	G	N	N	R	R	G	R
5.2	Landing with simulated jammed horizontal stabiliser in any out-of-trim position	TP	S	S	S	S	S	S	S	R	R	S	S	S	S	S
		T	R	R	R	R	R	R	R	G	N	R	R	N	G	R
5.3	Crosswind landings (aircraft, if practicable)	TP	S	S	S	S	S	S	S	R	R	S	S	N	R	R
		T	R	R	R	R	R	R	R	G	N	R	R	N	G	R
5.4	Traffic pattern and landing without extended or with partly extended flaps and slats	TP	S	S	S	S	S	S	S	R	R	S	S	N	R	R
		T	R	R	R	R	R	R	R	G	N	N	R	N	G	R
5.5	Landing with critical engine simulated inoperative	TP	S	S	S	S	S	S	S	R	R	S	S	N	R	R
		T	R	R	R	R	R	R	R	G	G	R	R	N	G	R
5.6	Landing with two engines inoperative: — aeroplanes with three engines: the centre engine and one outboard engine as far as practicable according to the data of the AFM; and — aeroplanes with four engines: two engines at one side	TP	S	S	S	S	S	S	S	R	R	S	S	S	S	S
		T	R	R	R	R	R	R	R	G	G	R	R	N	G	R

(ii) for helicopters:

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight controls forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
Section 1 — Flight preparation																
1.1	Helicopter exterior visual inspection; location of each item and purpose of inspection	TP	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		T	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.2	Cockpit inspection	TP	S	R	G	S	G	N	N	N	N	N	N	N	G	N
		T	R	R	G	R	G	N	N	N	N	N	N	N	G	N
1.3	Starting procedures, radio and navigation equipment check, selection and setting of navigation and communication frequencies	TP	S	S	S	S	S	N	N	R	R	R	G	S	R	G
		T	R	R	R	R	R	N	N	G	N	N	G	R	G	G
1.4	Taxiing/air taxiing in compliance with ATC instructions or with instructions of an instructor	TP	S	S	S	S	S	S	N	R	R	R	R	N	R	R
		T	R	R	R	R	R	R	N	G	N	N	R	N	G	G
1.5	Pre-take-off procedures and checks	TP	S	S	S	S	S	S	N	R	R	R	R	S	R	R
		T	R	R	R	R	R	R	N	G	N	N	R	R	G	G
Section 2 — Flight manoeuvres and procedures																
2.1	Take-offs (various profiles)	TP	S	S	S	S	S	S	S	R	G	R	S	N	R	R
		T	R	R	R	R	R	R	R	G	N	N	R	N	G	G
2.2	Sloping ground or crosswind take-offs & landings	TP	S	S	S	S	S	S	S	R	G	R	S	N	R	R
		T	R	R	R	R	R	R	R	G	N	N	R	N	G	G
2.3	Take-off at maximum take-off mass (actual or simulated maximum take-off mass)	TP	S	S	S	S	S	S	S	R	G	R	S	N	R	R
		T	R	R	R	R	R	R	R	G	N	N	R	N	G	G

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight controls forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
2.4	Take-off with simulated engine failure shortly before reaching TDP or DPATO	TP T	S R	S R	S R	S R	S R	S R	S R	R G	R N	R N	S R	N N	R G	R G
2.4.1	Take-off with simulated engine failure shortly after reaching TDP or DPATO	TP T	S R	S R	S R	S R	S R	S R	S R	R G	R N	R N	S R	N N	R G	R G
2.5	Climbing and descending turns to specified headings	TP T	S R	S R	S R	S R	N N	N N	S R	G G	G N	G N	G N	N N	R G	N N
2.5.1	Turns with 30° bank, 180° to 360° left and right, by sole reference to instruments	TP T	S R	S R	S R	S R	N N	N N	S R	G G	G N	G N	N N	N N	R G	N N
2.6	Autorotative descent	TP T	S R	S R	S R	S R	N N	N N	S R	R G	S N	S N	R G	N N	R G	G G
2.6.1	For single-engine helicopters (SEH), autorotative landing or for multi-engine helicopters (MEH) power recovery	TP T	S R	S R	S R	S R	S R	S R	S R	R G	S N	S N	S R	N N	R G	R G
2.7	Landings, various profiles	TP T	S R	S R	S R	S R	S R	S R	S R	R G	R N	R N	S R	N N	R G	R G
2.7.1	Go-around or landing following simulated engine failure before LDP or DPBL	TP T	S R	S R	S R	S R	S R	S R	S R	R G	R N	R N	S R	N N	R G	R G
2.7.2	Landing following simulated engine failure after LDP or DPBL	TP T	S R	S R	S R	S R	S R	S R	S R	R G	R N	R N	S R	N N	R G	R G
Section 3 — Normal and abnormal operations of the following systems and procedures																
3.1	Engine	TP T	S G	S R	S G	S S	S G	S G	S G	R G	G N	G N	G G	N N	G G	G G
3.2	Air conditioning (heating, ventilation)	TP	S	N	G	S	N	N	G	N	N	N	N	N	G	N

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight controls forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
		T	G	N	G	S	N	N	G	N	N	N	N	N	G	N
3.3	Pitot/static system	TP	S	N	G	S	N	N	G	N	N	N	G	N	R	N
		T	G	N	G	S	N	N	G	N	N	N	G	N	R	N
3.4	Fuel system	TP	S	N	G	S	N	N	G	N	N	N	N	N	G	N
		T	G	N	G	S	N	N	G	N	N	N	N	N	G	N
3.5	Electrical system	TP	S	N	G	S	G	G	G	N	N	N	G	N	G	G
		T	G	N	G	S	G	G	G	N	N	N	G	N	G	G
3.6	Hydraulic system	TP	S	S	S	S	R	R	R	N	N	N	G	N	G	G
		T	G	S	R	S	R	R	R	N	N	N	G	N	G	G
3.7	Flight control and trim system	TP	S	S	S	S	G	G	G	N	N	N	G	N	G	G
		T	G	S	R	S	G	G	G	N	N	N	G	N	G	G
3.8	Anti-icing and de-icing system	TP	R	N	G	R	N	N	G	G	G	G	N	N	R	N
		T	G	N	G	G	N	N	G	G	N	N	N	N	G	N
3.9	Autopilot/stability augmentation devices/flight director	TP	S	S	S	S	S	S	S	N	G	G	R	R	S	R
		T	G	R	R	R	R	R	R	N	N	N	G	R	R	R
3.10	EGPWS/HTAWS (if applicable)	TP	S	R	G	S	N	N	G	N	N	N	R	N	G	R
		T	G	G	G	R	N	N	G	N	N	N	G	N	G	R
3.11	Weather radar, radio altimeter, transponder	TP	S	N	G	S	N	N	G	N	N	N	G	N	R	R
		T	G	N	G	S	N	N	G	N	N	N	G	N	R	R
3.12	Area navigation system	TP	S	N	G	S	N	N	G	N	N	N	N	R	G	G
		T	G	N	G	S	N	N	G	N	N	N	N	R	G	G
3.13	Landing gear system	TP	S	R	R	S	R	R	R	N	G	G	R	N	G	R
		T	G	R	R	S	R	R	R	N	G	G	R	N	G	R

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight controls forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
3.14	APU	TP T	S G	N N	G G	S S	N N	N N	G G	N N	N N	N N	N N	N N	G G	N N
3.15	Radio, navigation equipment, instruments and FMS	TP T	S G	N N	G G	S S	N N	N N	G G	N N	N N	N N	N N	S R	G G	G G
Section 4 — Abnormal and emergency procedures																
4.1	Fire drills (including evacuation if applicable)	TP T	S S	S R	S R	S S	S R	S R	S R	N N	N N	N N	R R	N N	G G	R R
4.2	Smoke control and removal	TP T	S S	N N	G G	S S	N N	N N	G G	N N	N N	N N	R R	N N	G G	R R
4.3	Engine failures, shutdown and restart at a safe height	TP T	S R	S R	S R	S S	N N	S G	S G	R G	R G	R G	R G	G G	G G	R R
4.4	Fuel dumping (simulated, if applicable)	TP T	R G	N N	G G	R G	N N	N N	G G	N N	N N	N N	N N	N N	G G	N N
4.5	Tail rotor control failure (if applicable)	TP T	R G	S R	S R	S R	R R	S R	S R	N N	G G	S R	S R	N N	G G	R R
4.5.1	Tail rotor loss (if applicable)	TP T	R G	S R	S R	S R	R R	S R	S R	R N	S G	S R	S R	N N	G G	R R
4.6	Incapacitation of crew member – only for MPH and SPH operated in MPO	TP T	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -
4.7	Transmission malfunctions	TP T	S R	S R	S R	S R	R R	R R	R R	G G	G G	G G	R G	N N	G G	G G
4.8	TCAS event (if applicable)	TP T	S G	R G	G G	S R	N N	N N	G G	N N	N N	N N	R N	N N	G G	N N

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight controls forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
4.9	Other emergency procedures as outlined in the appropriate flight manual	TP T	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -
Section 5 — Instrument flight procedures																
5.1	Instrument take-off: transition to instrument flight is required as soon as possible after becoming airborne	TP T	S R	S R	S R	S R	N N	S R	S R	G G	R N	R N	R R	S R	R G	G G
5.1.1	Simulated engine failure during departure	TP T	S R	S R	S R	S R	N N	S R	S R	G G	R N	R N	R G	S R	R G	G G
5.2	Adherence to departure and arrival routes and ATC instructions	TP T	R G	R G	R G	S R	N N	N G	R G	G G	N N	N N	N R	S R	R G	N N
5.3	Holding procedures	TP T	R G	R G	R G	S R	N N	N G	R G	G G	G N	G N	N R	S R	R G	N N
5.4.1	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach, manually, without flight director	TP T	S R	S R	S R	S R	N N	N G	R G	G G	N N	N G	G R	S R	R G	G G
5.4.2	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach, manually, with flight director	TP T	S R	S R	S R	S R	N N	N G	R G	G G	N N	N G	G R	S R	R G	G G
5.4.3	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach, with coupled autopilot	TP T	S R	R G	R G	S R	G G	R G	R G	G G	N N	N G	G R	S R	R G	G G
5.4.4	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach, manually, with one engine simulated inoperative; engine failure has to be	TP T	S R	S R	S R	S R	N N	N G	R G	G G	R N	R N	S R	S R	R G	R G

	Manoeuvres/Procedures	Training to Proficiency (TP) / Training (T)	1. Flight deck layout and structure	2. Flight controls forces and hardware	3. Flight controls systems operation	4. Aircraft systems	5. Performance and handling on ground	6. Performance and handling in ground effect	7. Performance and handling out of ground effect	8. Sound cueing	9. Vibration cueing	10. Motion cueing	11. Visual cueing	12. Navigation	13. Atmosphere and weather	14. Operating sites and terrain
	simulated during final approach before passing 1 000 ft above aerodrome level until touchdown or until completion of the missed approach procedure															
5.5	2D operations down to the MDA/H	TP	S	S	S	S	N	N	R	G	N	N	G	S	R	G
		T	R	R	R	R	N	N	G	G	N	N	G	R	G	G
5.6	Go-around with all engines operating on reaching DA/H or MDA/MDH	TP	S	S	S	S	N	N	R	G	R	R	G	S	R	G
		T	R	R	R	R	N	N	G	G	N	N	G	R	G	G
5.6.1	Other missed approach procedures	TP	S	S	S	S	N	N	R	G	R	R	G	S	R	G
		T	R	R	R	R	N	N	G	G	N	N	G	R	G	G
5.6.2	Go-around with one engine simulated inoperative on reaching DA/H or MDA/MDH	TP	S	S	S	S	N	N	R	G	R	R	G	S	R	G
		T	R	R	R	R	N	N	G	G	N	N	G	R	G	G
5.7	IMC autorotation with power recovery	TP	S	S	S	S	N	N	R	G	R	R	R	N	G	N
		T	R	R	R	R	N	N	G	G	N	N	R	N	G	N
5.8	Recovery from unusual attitudes manually and, if applicable, with auto recovery mode	TP	S	S	S	S	N	N	R	G	R	R	N	N	G	N
		T	R	R	R	R	N	N	G	G	N	N	N	N	G	N

(b) Training considerations

- (1) The training matrices have been created by considering the type rating training course requirements in accordance with Part-FCL. When training is provided in accordance with Regulation (EU) No 965/2012 for operations that require specific approval, a careful evaluation of the fidelity levels required for the relevant features of an FSTD to be used for certain training tasks should be carried out, taking into consideration the technical specifications of the training device included in the qualification certificate and in the ESL.

Detailed information regarding the general requirements for each FSTD feature can be found in Subpart B of CS-FSTD Issue 1.

- (2) At TP level, in general, the 'Flight deck layout and structure' and the 'Flight controls forces and hardware' features should have tactile hardware for the relevant aircraft systems and equipment to be operated during the execution of the relevant training tasks, unless the specific aircraft type is equipped with touchscreen devices for specific aircraft systems. However, for the 'Flight controls forces and hardware' and 'Flight deck layout and structure' features, a 'Representative' (R) fidelity level is indicated at TP level for some training tasks for which a touchscreen representation may be allowed according to CS-FSTD Issue 1. If where those training tasks are completed at TP level in FSTDs whose aforementioned features consist of touchscreen representation, elements of those training tasks should additionally be completed in FSTDs that have tactile hardware.

At T level, the 'Flight deck layout and structure' and the 'Flight controls forces and hardware' features may be a touchscreen representation, except for the aircraft systems and equipment which need to be directly operated when flying manually.

- (3) In the case of training tasks or subtasks that have an influence on, or that should be executed in conjunction with, other training tasks, the FSTD capability signatures (FCSs) indicated in the training matrix for such training tasks should be combined by taking, for each training task, the highest fidelity level for each feature.

- (4) For aeroplane training tasks included in Section 3.4 and Section 3.6 of the table in point (a)(3)(i), and for helicopter training tasks included in Section 3 and Section 4 of the table in point (a)(3)(ii), the fidelity level of the relevant flight control hardware and/or aircraft systems of the FSTD for the 'Flight controls forces and hardware' and 'Aircraft systems' features should match the fidelity level indicated in the training matrix. The ESL should be used as the reference document to find out the fidelity level information for the relevant flight control hardware and/or aircraft systems.

For FSTDs with assigned FCS, the fidelity level of 'Flight controls forces and hardware' and 'Aircraft systems' features in the FSTD qualification certificate may overestimate the real representativeness of some flight control forces or hardware, or of some aircraft systems. For this reason, the information included in the ESL regarding the representativeness of flight control forces and hardware and aircraft systems should be used, in addition to assessing the suitability of the FSTD for training the relevant training tasks in accordance with point ORA.ATO.135(b)(1) of Annex VII (Part-ORA) of Regulation (EU) No 1178/2011.

When an FSTD has differences in fidelity levels across flight deck representation of aircraft system interfaces or any omitted/non-interactive system interfaces for the 'Flight deck layout and structure' feature, the use of such FSTD for a particular task is determined based on the assessment of the adequacy of aircraft system interfaces and their fidelity level and the training tasks. Such assessment should be performed in accordance with point ORA.ATO.135(b)(1) of Annex VII (Part-ORA) of Regulation (EU) No 1178/2011.

- (5) At T level, the FCSs for aeroplane training tasks included in Section 3.4 and Section 3.6 of the table in point (a)(3)(i) show a 'Representative' (R) fidelity for the 'Aircraft systems' feature. If an aircraft type shows significant differences between variants for certain aircraft systems, training providers should consider using an FSTD that has a 'Specific' (S) fidelity level for the 'Aircraft systems' feature, when providing training for a particular variant.
- (6) For the aeroplane training tasks 3.6.7 and 3.6.8 of the table in point (a)(3)(i) or the helicopter training tasks 4.6 and 4.9 of the table in point (a)(3)(ii), the suitability of the FSTD should be determined based on the assessment in accordance with point ORA.ATO.135(b)(1) of Annex VII (Part-ORA) of Regulation (EU) No 1178/2011.
- (7) Except for aeroplane training task 2.2 of the table in point (a)(3)(i), take-off and landing training tasks have a 'None' (N) fidelity level for the 'Navigation' feature, as take-offs and landings are performed visually. However, in most cases, especially at TP level, such training tasks are followed by instrument departures or are executed following an instrument arrival and approach procedure. In such cases, when combining different training tasks as per point (b)(3), the FCSs indicated in the training matrices for take-off and landing training tasks should be combined with those of the relevant instrument departure or instrument arrival and approach procedures, and the highest resulting fidelity level for the 'Navigation' feature should be taken.
- (8) In general, motion and vibration cues should not be required for instrument flight procedures, as it is assumed that the scope of the training tasks is focused on the correct execution of the instrument flight procedures and not on the recognition of cueing experienced by the flight crew. However, for specific training tasks which require dynamic manoeuvring and significant changes of aircraft attitude, such as, for aeroplanes, one-engine simulated inoperative procedures, or circling and visual approaches, motion cues and, where needed, vibration cues, should be required.

GM1 to Appendix 9 Training, skill test and proficiency check for the MPL, and the ATPL, and for type and class ratings, and proficiency checks for the BIR and the IR

TYPE SPECIFIC UPRT AND GO-AROUND TRAINING IN FSTD

[...]

- (b) Stall event recovery in FSTD (Appendix 9, Section B(5) exercise 7.2.1; Section B(6) exercise 3.7.1)
 - (1) It is of utmost importance that stall event recovery training takes into account the capabilities of the FFS used. To deliver stall event recovery training, the FFS should be

qualified against the relevant UPRT elements of CS-FSTD(A) Issue 2 or CS-FSTD Issue 1. Stall event recovery training should include training up to the stall (approach-to-stall). Post-stall training may be delivered provided the device has been qualified against the relevant optional elements of CS-FSTD(A) Issue 2 or CS-FSTD Issue 1 and the operator demonstrates that negative training or negative transfer of training is avoided. A 'stall event' is defined as an occurrence whereby the aeroplane experiences one or more conditions associated with an approach to-stall or a post stall.

[...]

[...]

GM1 to Appendix 9 Section A point 1f Training, skill test and proficiency check for the MPL, and the ATPL, and for type and class ratings, and proficiency checks for the BIR and the IR

FSTD FEATURES AND FIDELITY LEVELS

- (a) This guidance material provides information on the FSTD features and fidelity levels of an FCS.
- (b) An 'FSTD simulation feature' refers to a domain of simulation for which CS-FSTD establishes certification specifications. Fourteen FSTD features are grouped into three categories as follows:
 - (1) Aircraft simulation features
 - 1. Flight deck layout and structure (FDK)
 - 2. Flight controls forces and hardware (CLH)
 - 3. Flight controls systems operation (CLO)
 - 4. Aircraft systems (SYS)
 - 5. Performance and handling on ground (GND)
 - 6. Performance and handling in ground effect (IGE)
 - 7. Performance and handling out of ground effect (OGE)
 - (2) Cueing simulation features
 - 8. Sound cueing (SND)
 - 9. Vibration cueing (VIB)
 - 10. Motion cueing (MNT)
 - 11. Visual cueing (VIS)
 - (3) Environment simulation features
 - 12. Navigation (NAV)
 - 13. Atmosphere and weather (ATM)
 - 14. Operating sites and terrain (OST)

(c) A 'fidelity level' indicates the extent to which an FSTD represents an aircraft. Four fidelity levels are used as follows:

(1) Specific (S)

The highest level of fidelity for a given FSTD simulation feature. The feature replicates a particular variant of an aircraft type and/or specific operating environment.

(2) Representative (R)

The intermediate level of fidelity for a given FSTD simulation feature. The feature has the characteristics of an aircraft type and/or representative operating environment.

(3) Generic (G)

The lowest level of fidelity for a given FSTD simulation feature. The feature has the characteristics of a group of aircraft and/or generic operating environment.

(4) None (N)

The feature that is not installed, functional or available for use in training.

(d) An FSTD simulation feature category referred to in point (b) is considered to correspond to a fidelity level referred to in point (c) in accordance with the following descriptions:

(1) S = Specific

(i) Aircraft simulation feature

Replicates the appearance, tactile feel, system operation, performance, and handling of the simulated aircraft type and variant. Highest level of the simulation model fidelity.

(ii) Cueing simulation feature

The highest level of cueing requiring the most realistic simulation for the cueing.

(iii) Environment simulation feature

Replicates the real-world environment for specified locations.

The highest level of environment simulation requiring the most realistic simulation.

(2) R = Representative

(i) Aircraft simulation feature

Is characteristic of the appearance, tactile feel, system operation, performance, and handling of the simulated aircraft type. Simulation may include elements from different variants of the same aircraft type.

(ii) Cueing simulation feature

The intermediate level of cueing.

(iii) Environment simulation feature

The intermediate level of environment simulation.

(3) G = Generic

(i) Aircraft simulation feature

Is characteristic of the appearance, tactile feel, system operation, performance, and handling of the simulated aircraft group. Simulation may include elements from different aircraft of the same aircraft group. Lowest level of the simulation model fidelity.

(ii) Cueing simulation feature

The lowest level of cueing.

(iii) Environment simulation feature

The lowest level of environment simulation.

(4) N = None

(i) All

Features are not evaluated. If a feature is installed, but is not required, it should not be distracting or detract from the other features for the intended use of the device.

(e) Further information regarding the description of the features and fidelity levels can be found in Subpart B of CS-FSTD Issue 1.

GM1 to Appendix 9 Section A point 1g Training, skill test and proficiency check for the MPL and the ATPL, and for type and class ratings, and proficiency check for the BIR and the IR

USE OF OTDs IN TYPE RATING TRAINING COURSES AND TRAINING TIME

OTDs may be used in type rating training courses where so permitted by Appendix 9 to Annex I (Part-FCL). The minimum FSTD training time for a type rating training course is either determined in point (j) of AMC2 ORA.ATO.125 for aeroplanes or in AMC2 FCL.725(a) for helicopters, unless otherwise foreseen in the OSD.

As OTDs are non-qualified training devices, the training time spent using such devices should not be counted towards the minimum FSTD time. However, such time constitutes part of the overall type rating training course time.